Solderbench: An open-source synthetic benchmark for PCB solder-joint defect inspection algorithm development

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Abstract

Automated optical inspection (AOI) of solder joints is a critical quality-control step in printed circuit board (PCB) assembly, and three-dimensional (3D) point-cloud inspection is increasingly adopted because solder quality is essentially a volumetric property. However, publicly available 3D solder-joint datasets are essentially nonexistent: industrial scan data are proprietary, so new inspection algorithms are difficult to compare fairly. We present Solderbench, an open-source, pure-Python synthetic benchmark for developing and evaluating 3D solder-joint defect inspection algorithms. Solderbench provides (i) a parametric point-cloud generator that simulates five joint classes (good, insufficient, excessive, bridging, and collapse) with more than fifteen controllable parameters and seed-level, point-wise reproducibility; (ii) a seven-dimensional geometric feature extractor; (iii) six baseline classifiers, including a re-implementation of the industrial volume-threshold rule S3DA; and (iv) automated comparison and ablation runners that export CSV, LaTeX, and figure artifacts. On a 1250-sample benchmark (seed 42), a random forest reaches 0.9440 accuracy and 0.9439 macro-F1, and feature ablation identifies height variance as the most informative descriptor. Synthetic results provide a controlled, reproducible proxy for algorithm development, but should ultimately be validated on real production scans.

Keywords: solder joint inspection; automated optical inspection; 3D point cloud; synthetic benchmark; defect classification; reproducible evaluation; PCB; synthetic dataset

1. Required metadata

2. Motivation and significance

SMT dominates modern printed circuit board (PCB) assembly, and solder-joint quality largely determines product reliability. AOI is the standard quality-control step on SMT lines [1, 2, 3]; classical 2D AOI and deep-learning 2D detectors on public datasets such as DeepPCB [4, 5] are mature, yet inspection is increasingly moving to 3D because solder quality is fundamentally volumetric [6, 7]. General 3D deep learning has advanced rapidly – with PointNet and PointNet++ [8, 9] and tooling such as Open3D [10] – but 3D solder-joint research remains hampered by the absence of public data: existing public PCB datasets are almost entirely 2D [5], and the few 3D studies rely on private, non-downloadable in-house point clouds [6, 7], so algorithms cannot be compared fairly or reproduced. Solderbench closes this gap synthetically, simulating solder-joint point clouds from a parametric geometric model whose parameters (nominal joint size, noise, defect strength, bridging distance, collapse depth) are physically interpretable and user-controllable; synthetic generation yields costless labelled data of arbitrary size, deliberate control of class difficulty and sensor noise, and seed-level reproducibility.

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Table 1: Code metadata (mandatory).

Nr.	Code metadata description	Value
C1	Current code version	v0.1.0
C2	Permanent link to code/repository used for this code version	https://github.com/OWNER/solderbench
C3	Permanent link to reproducible capsule	10.5281/zenodo.XXXXXXX
C4	Legal code license	MIT License
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python ($\geq$ 3.8)
C7	Compilation requirements, operating environments and dependencies	numpy, scipy, pandas, scikit-learn, matplotlib; OS-independent (no compilation, no GPU required)
C8	Link to developer documentation/manual	https://github.com/OWNER/solderbench#readme
C9	Support email for questions	wangzz@gdut.edu.cn

3. Software description

3.1. Software architecture

Solderbench is a pure-Python package (about 1,000 source lines) with five feed-forward modules behind one public API: `generator.py` (parametric simulator), `features.py` (seven descriptors), `baselines.py` (six classifiers), `evaluate.py` (`run_comparison`/`run_ablation` runners exporting CSV, booktabs LaTeX, and figures), and `data.py` (`load_dataset()`, the real-camera integration point). Stages exchange NumPy arrays / pandas DataFrames, so any stage is replaceable; synthetic clouds swap for segmented real scans at `load_dataset()` without downstream changes. The package ships a 15-test suite (shape/balance, same-seed reproducibility, feature validity, a Welch t-test that insufficient-solder volumes are significantly smaller, and fit/predict smoke tests), a `pyproject.toml`, and a `codemeta.json`; dependencies are only numpy, scipy, pandas, scikit-learn, and matplotlib [14, 15, 16, 17].

3.2. Software functionalities

Generation. SolderJointGenerator simulates five SMT defect classes [6, 7]: good, insufficient (shrunk ellipsoid, 0.55–0.85 of nominal), excessive (inflated ellipsoid, mean inflation 1.3, plus a bump), bridging (small centre-to-centre distance plus a strip), and collapse (random top dent). Over fifteen parameters control n_{points}, noise, and the shrink/inflation/bridging/collapse ranges; class ranges deliberately overlap so no classifier is perfect, and difficulty is tuned by widening or narrowing them.

Features. extract_features returns seven descriptors (Table 2): `vol` (voxel volume), `max_h`, `h_var`, `area`, `vol_area_ratio`, `convex_vol_ratio` (irregularity), and `min_dist` (bridging exposure). **Baselines and**

Table 2: The seven geometric descriptors returned by `extract_features` and their physical meaning.

Feature	Definition	Physical meaning
`vol`	voxel count $\times$ (voxel size)^3	total solder volume
`max_h`	max z over points	joint height
`h_var`	variance of point heights	height-distribution spread (dent / bump)
`area`	convex-hull surface area	joint surface extent
`vol_area_ratio`	vol / area	compactness
`convex_vol_ratio`	vol / ConvexHull.volume	shape irregularity (voids / dents)
`min_dist`	min distance to neighbouring joint	bridging exposure

runners. get_baselines returns S3DA – a re-implementation of the industrial volume-threshold rule [6], threshold fixed in source and tunable – plus logistic regression, an RBF-SVM [12], kNN, gradient boosting, and a random forest [11]; all are scikit-learn estimators [13] with fixed random states. run_comparison(X, y) evaluates all six under a stratified 7:3 split with 5-fold CV, returning accuracy, macro precision/recall/F1, and CV accuracy; run_ablation(X, y) does leave-one-feature-out and single-feature analysis anchored on the random forest (Table 3, B6) to isolate descriptor contribution. Both export CSV and LaTeX, run_comparison also renders the best model's confusion matrix, and `python examples/run_benchmark.py` reproduces the full benchmark.

Validity guarantees. Every cloud is resampled to exactly n_{points} points, preventing sampling-density leakage so volumetric features reflect shape rather than density. All randomness derives from one independent `RandomState` created at construction (not NumPy's global seed), so a given seed yields point-wise identical clouds, features, and tables – asserted by the test suite. There is no compiled code, GPU, or network access at runtime.

4. Illustrative examples

4.1. Quickstart

Install the package and its dependencies first with `pip install -e .` (requires Python $\geq$ 3.8); the API below corresponds to version v0.1.0 and may change in later releases. The following end-to-end example generates a dataset, extracts features, and runs both experiments in eight lines:

```
from solderbench import (SolderJointGenerator, extract_features,
get_baselines, run_comparison, run_ablation)
# generate 1250 joints (5 classes x 250) at seed 42
gen = SolderJointGenerator(n_points=1500, noise=0.02, seed=42)
clouds, labels = gen.generate(n_per_class=250)   # 5 classes x 250 joints
X, y = gen.to_features(clouds, labels)           # 7-feature matrix + labels
results = run_comparison(X, y)   # 6 methods x Acc/P/R/F1 + 5-fold CV
print(results[["no", "name", "Accuracy", "F1"]])
abl = run_ablation(X, y)         # leave-one-out + single-feature dF1
print(abl.head(8))
```

4.2. Benchmark comparison

`examples/run_benchmark.py` produces the benchmark snapshot reported in Table 3: 1250 samples (5 classes $\times$ 250 joints, 1500 points per cloud, seed 42, stratified 7:3 split; 5-fold cross-validation on the training set). Three observations follow. First, the industrial rule S3DA (B1) is a strong baseline at 0.8960 accuracy, confirming

Table 3: Baseline comparison on the Solderbench benchmark (seed = 42, 1250 samples). Best results in bold.

No.	Method	Accuracy	Precision	Recall	macro-F1	CV Acc. (5-fold)
B1	S3DA (volume threshold)	0.8960	0.8990	0.8960	0.8964	0.9234 ± 0.0138
B2	Logistic Regression	0.9387	0.9416	0.9387	0.9386	0.9657 ± 0.0198
B3	SVM (RBF)	0.9387	0.9390	0.9387	0.9385	0.9771 ± 0.0177
B4	KNN	0.8933	0.8944	0.8933	0.8921	0.9234 ± 0.0168
B5	GradientBoosting	0.9413	0.9417	0.9413	0.9412	0.9577 ± 0.0138
B6	**Random Forest**	**0.9440**	**0.9452**	**0.9440**	**0.9439**	$\mathbf{0.9680 \pm 0.0160}$

that solder volume alone carries most of the class signal – exactly the physics the generator encodes. Second, all learning-based methods improve on the rule, with the random forest best on the held-out test set (accuracy 0.9440, macro-F1 0.9439). Third, test-set and cross-validation rankings do not fully coincide: the RBF SVM has the highest CV accuracy (0.9771) yet only 0.9387 on the held-out test set, a textbook signature of mild overfitting to the training folds. This is exactly why Solderbench reports both protocols and why users must trust the independent test set rather than cross-validation alone. Figure 1 shows the confusion matrix of the random forest on the held-out test set. Bridging is perfectly separated (75/75), because the `min_dist` feature exposes the abnormally small inter-joint distance by construction. The dominant residual confusion concentrates on the collapse class: 4 of its samples fall in good and 6 in insufficient, and a small number of good joints are also misclassified as collapse (3 samples). The

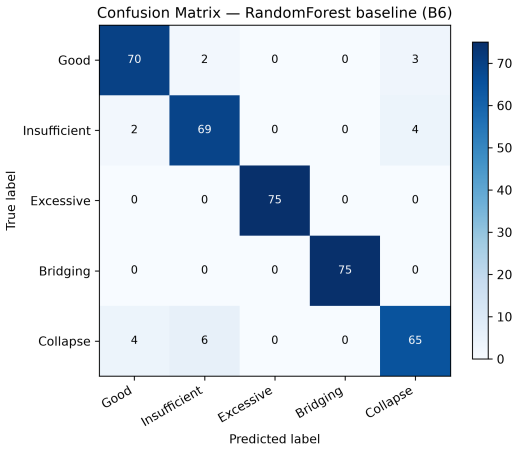


Figure 1: Confusion matrix for the Random Forest (B6) on the held-out test set (375 samples, seed = 42). Bridging is perfectly classified via the `min_dist` feature; the main confusion occurs between collapse and good/insufficient classes.

overall error therefore remains between collapse and the good/insufficient classes, because a shallow dent alters the height distribution only subtly – precisely why `h_var` – the most informative descriptor in the ablation (Table 4) – is needed to separate it; this mirrors the hard case reported for real 3D inspection systems [6, 7].

4.3. Feature ablation

Table 4 reports the leave-one-feature-out (LOO) ablation for the random forest (full-model macro-F1 = 0.9439). The ablation has a clean physical reading. Height variance (`h_var`) is the single most valuable descriptor because

Table 4: Leave-one-feature-out ablation of the RandomForest model (macro-F1; full model = 0.9439).

Removed feature	**dF1 vs. full model**	**Interpretation**
h_var	-0.0459	Most informative: collapse dents and excessive bumps reshape the height distribution
vol_area_ratio	-0.0105	Compactness cue separating volumetric defects from nominal joints
min_dist	-0.0054	Small overall cost because bridging is nearly separable already
area	-0.0052	Correlated with vol; adds modest independent signal
convex_vol_ratio	-0.0025	Irregularity cue; partially redundant with h_var
vol	+0.0026	Slightly redundant once ratios are present
max_h	0.0000	Fully redundant given h_var and vol

the two hardest defects – collapse (a dent) and excessive solder (a bump) – are primarily height-distribution phenomena, whereas volume-only cues saturate. Single-feature runs corroborate this: the strongest single feature is `convex_vol_ratio` (macro-F1 $= 0.6412$), which captures shape irregularity common to several defect types, while the weakest is `min_dist` (macro-F1 $= 0.3616$), which is highly informative for bridging but carries no signal for the other four classes. The near-zero or positive deltas for `vol` and `max_h` indicate information redundancy rather than irrelevance: both are strongly correlated with the ratio features. Volume alone is in fact a strong single cue – it underpins the industrial S3DA rule (B1), which reaches 0.8960 accuracy using volume only – and its near-zero delta here reflects redundancy with the ratio features, not uselessness. Removing `vol` slightly increases macro-F1 within small numerical fluctuations; this is purely a statistical artefact given that volume information is already encoded within ratio-based descriptors.

Table 5: Selected representative single-feature macro-F1 scores (RandomForest, seed = 42). Full results are programmatically exported by run_ablation.

Single feature	macro-F1
`convex_vol_ratio`	0.6412
`min_dist`	0.3616

5. Impact

Solderbench contributes to three communities. For inspection-algorithm research, it is the first public, reproducible 3D benchmark for solder-joint defect classification, replacing per-paper private datasets with a common task, protocol, and versioned reference numbers; new methods are ranked against six baselines with one command. For teaching, it is a self-contained geometric-ML case study that fits a short lab session and runs on any laptop, with each design decision doubling as a benchmark-methodology lesson. For reproducibility, seed-level determinism, a one-command script, machine-readable exports, a `codemeta.json` descriptor, and an MIT license make the software trivially citable, archivable, and reusable. The path to production data is short: a user with a real 3D AOI camera replaces the body of `load_dataset` – or passes their own X, y – with code that reads scans, segments joints, and calls `extract_features`; all downstream experiments run unchanged. Beyond the classical-ML workflow, the generator's raw point clouds can be passed directly to external deep models (e.g., PointNet [8, 9]); Solderbench supports both scenarios.

6. Limitations

We state the limitations of Solderbench honestly; they define the scope in which its results should be interpreted. The generator is a geometric model, not a physics simulator: it does not model wetting dynamics, surface tension, oxidation, or realistic sensor artifacts (occlusion, multi-path reflections, structured-light failure), and its noise is a simple isotropic perturbation; class distributions are plausible and overlapping, not fitted to production data. Three scope limits remain. First, the five classes are idealized single-fault geometries – compound defects (e.g., insufficient and collapsed together) are unsupported, although they occur on real lines. Second, the clouds model only the individual joint, omitting substrate, pads, and pins, so board-level context is out of scope. Third, resampling every cloud to a fixed n_{points} simplifies real sensors, which return a variable count, and may alter the point-density distribution. **Consequently, Solderbench is suitable for algorithm prototyping, fair cross-method comparison, and descriptor-level analysis, but its metric values must not be read as predictions of real industrial AOI performance.** Synthetic validation cannot substitute for validation on real scans; the benchmark guarantees only that comparisons made on it are fair, controlled, and exactly reproducible – the property the field currently lacks.

Data availability. The Solderbench source code and all reproduction scripts are released under the MIT license and archived on Zenodo (doi:`10.5281/zenodo.XXXXXXX`); the permanent GitHub repository is `https://github.com/OWNER/solderbench`. The benchmark dataset is not stored as a fixed file but generated deterministically at runtime by running `python examples/run_benchmark.py`, which regenerates every table and figure in this paper.

7. Conclusions

We presented Solderbench, an open-source, pure-Python synthetic benchmark for 3D PCB solder-joint defect inspection with a parametric five-class generator, seed-level reproducibility, a seven-dimensional feature extractor, six baselines including the industrial S3DA rule, and automated comparison/ablation runners. On the reference benchmark (1250 samples, seed 42) a random forest reaches 0.9440 accuracy and 0.9439 macro-F1, and ablation identifies height variance as the dominant descriptor. By removing the data-access barrier that fragments 3D solder-joint inspection, Solderbench makes algorithm development, comparison, and teaching open and reproducible. Future work: (i) fitting generator parameters to measured scans; (ii) adding sensor-artifact models; (iii) compound-defect taxonomy; (iv) real-data conversion scripts; (v) deep point-cloud model interfaces (e.g. PointNet, PointNet++); and (vi) preset difficulty templates.

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